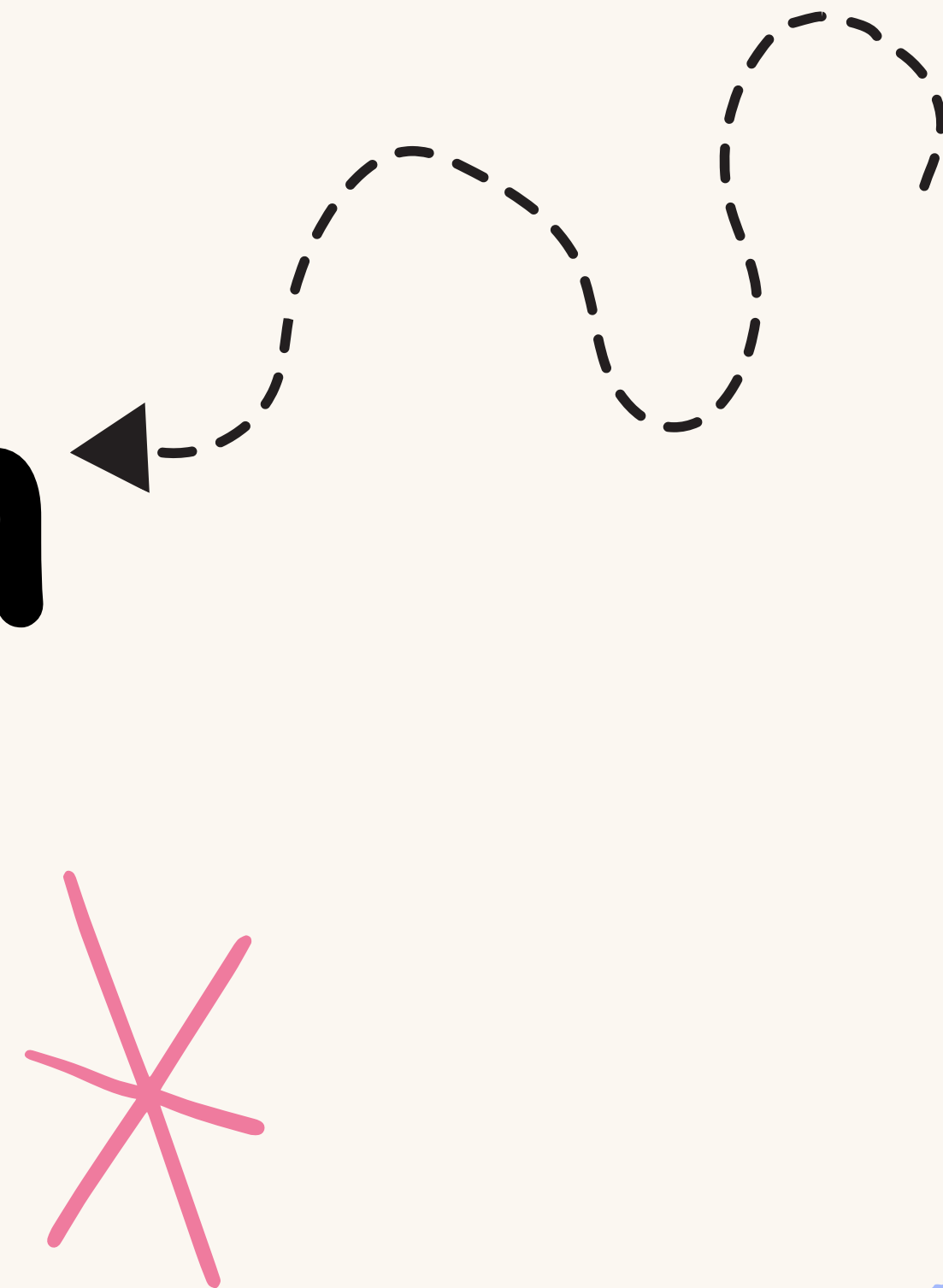
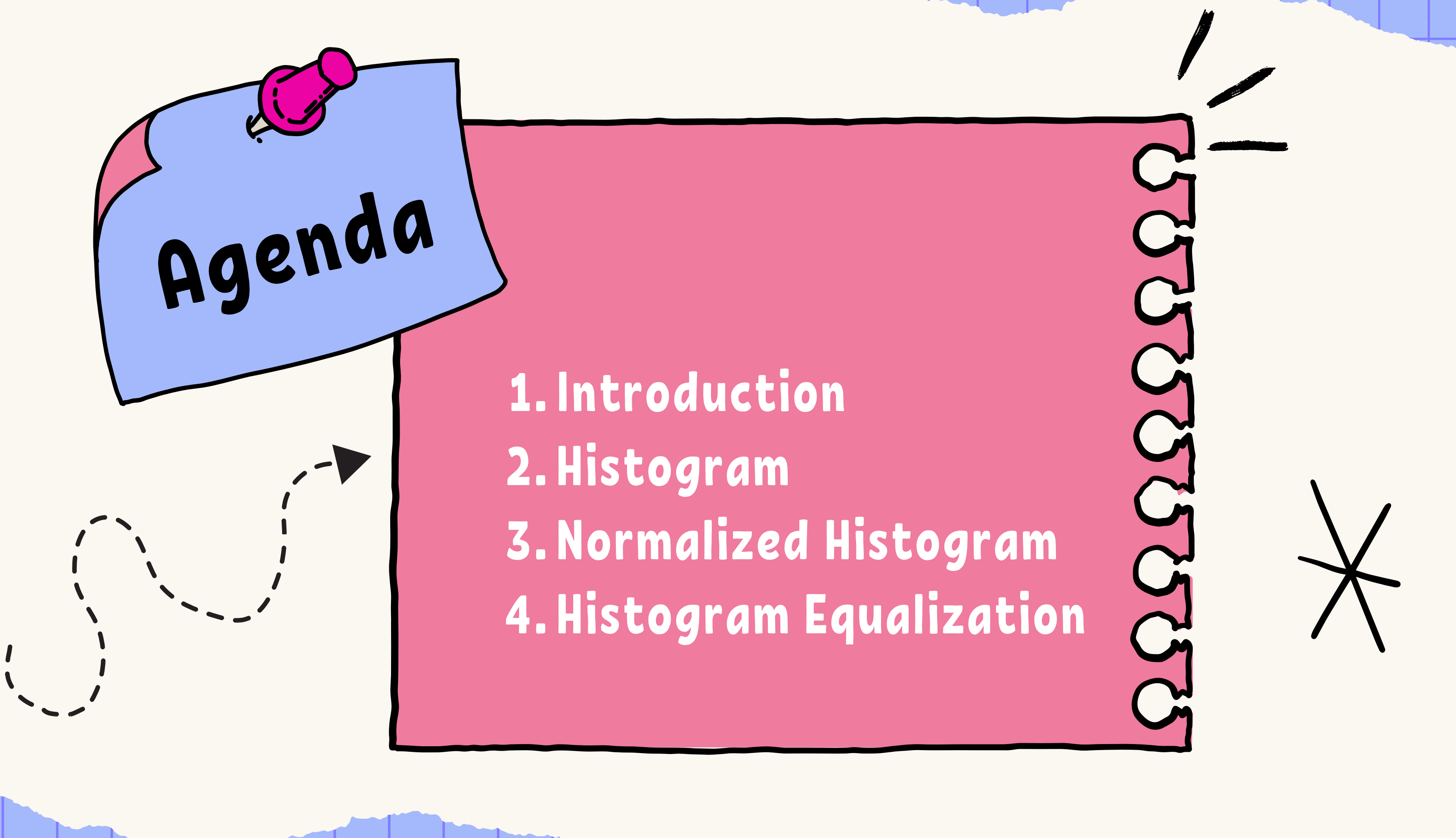


Histogram

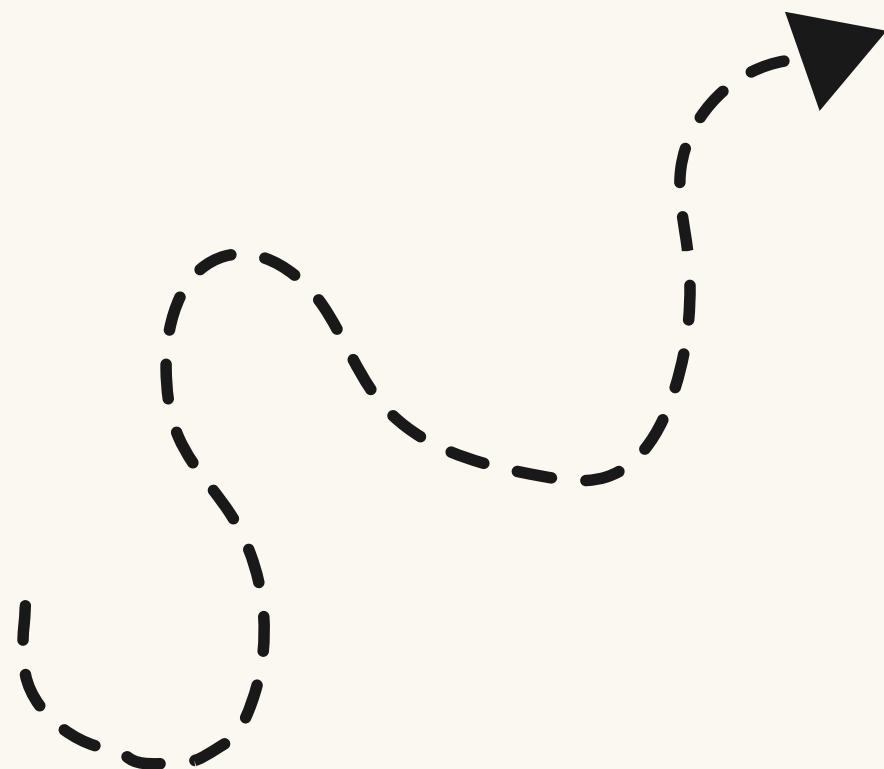
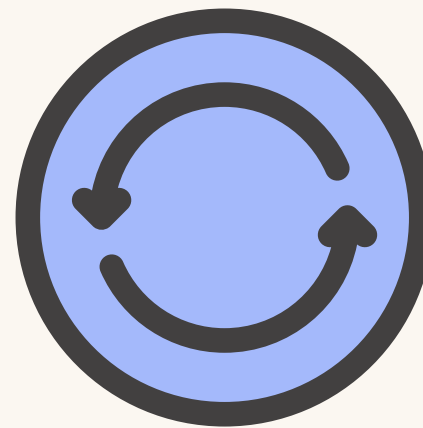




Agenda

1. Introduction
2. Histogram
3. Normalized Histogram
4. Histogram Equalization

**Let's introduce
some knowledge
first**

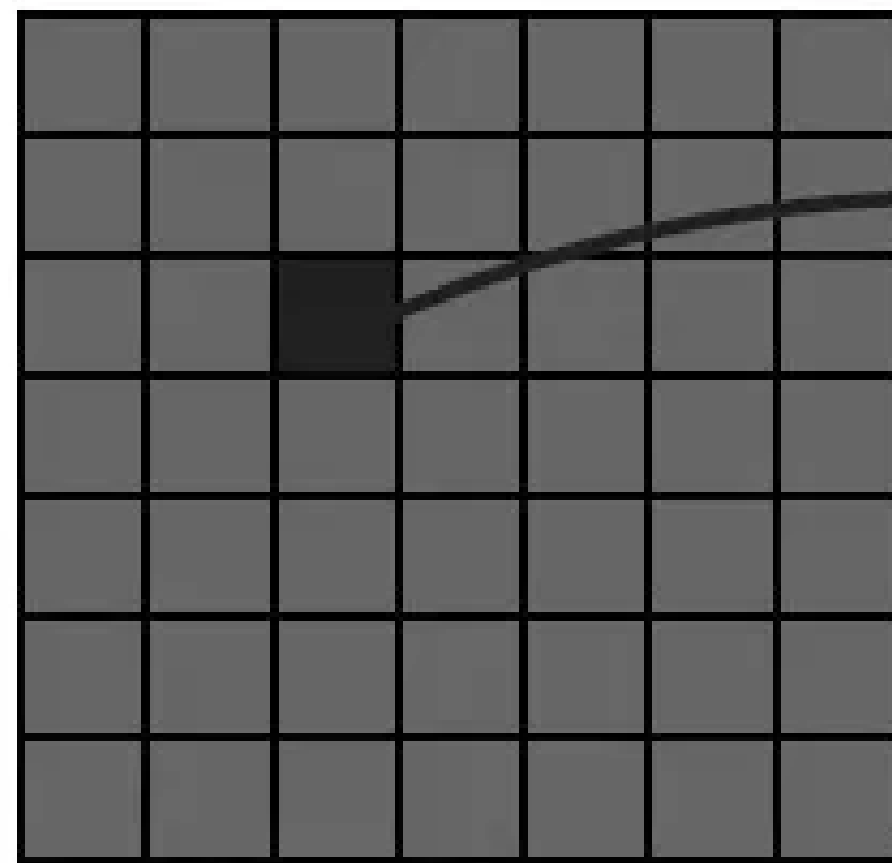


What is a Pixel Intensity ?

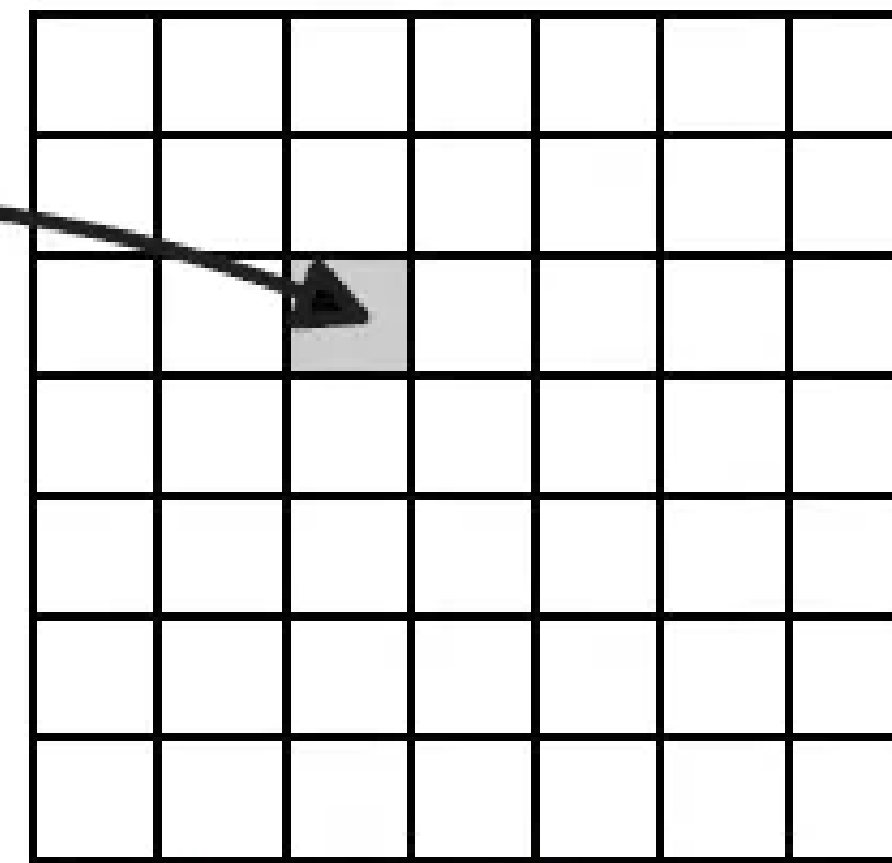
- **Each pixel in a grayscale image has a value representing its brightness, ranging from 0 (black) to 255 (white) for an 8-bit image.**
- **In color images, each pixel has three intensity values for Red (R), Green (G), and Blue (B) channels.**

What is Global Operation ?

refer to technique that process an entire image as a whole rather than focusing on specific pixels, regions, or neighborhoods



Input image



Output image

What is a Histogram in Image Processing?

A histogram is a graphical representation that shows the distribution of pixel intensities (brightness levels) in an image

A histogram plots these pixel values on the x-axis (intensity values) and the frequency of each intensity value on the y-axis (number of pixels with that intensity)

What is a Histogram in Image Processing?

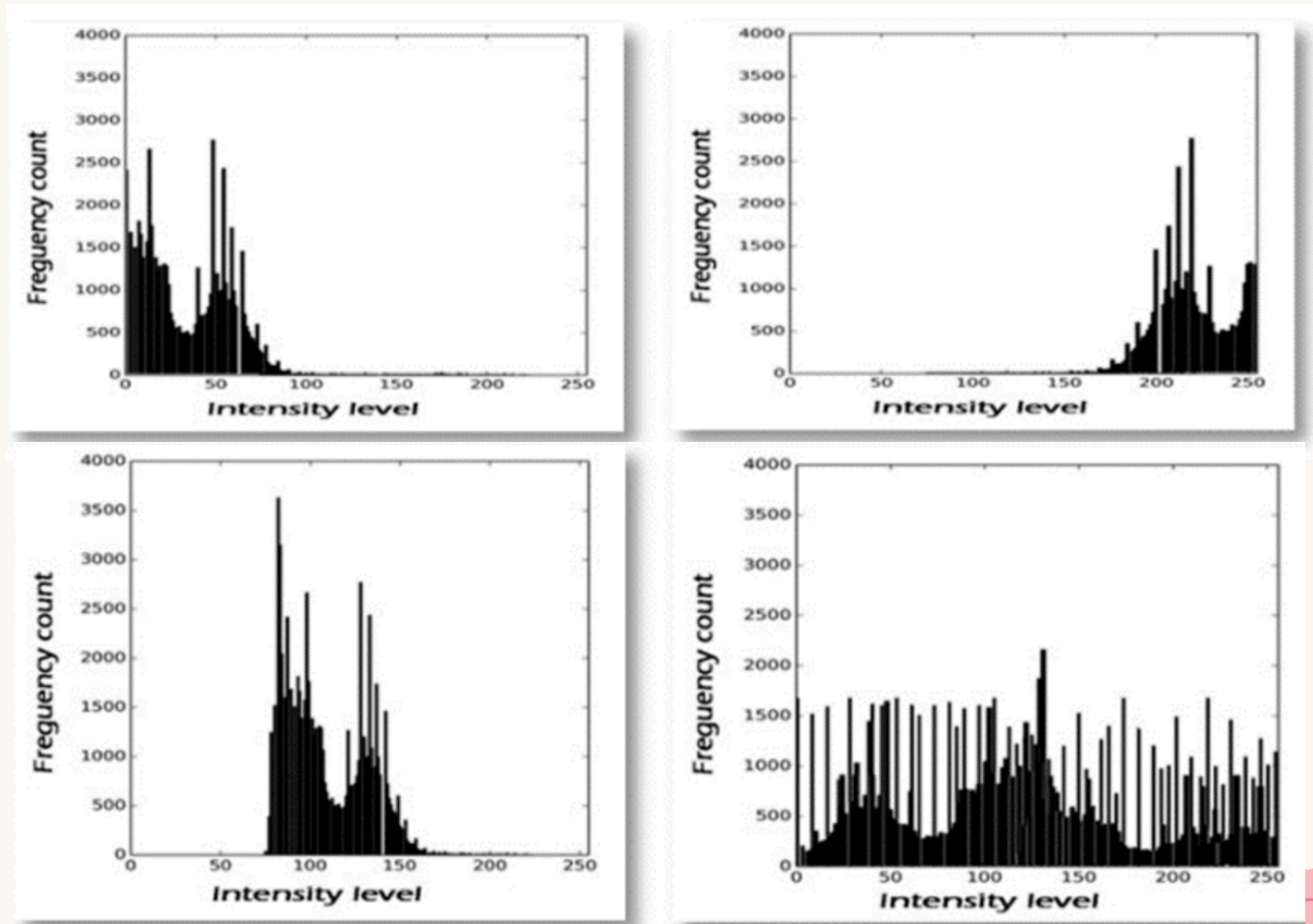
The histogram of a digital image with intensity levels in the range $[0: L-1]$ is a discrete function

$$h(r_k) = n_k$$

Where r_k is the k^{th} intensity value

n_k is the number of pixels with intensity r_k

Look at these Histograms 👁👁



What Does a Histogram Tell Us?

Image Characteristics:

- A dark image has more pixels with low intensity, so the histogram will be skewed to the left.
- A bright image has more pixels with high intensity, so the histogram will be skewed to the right.
- A high-contrast image has a wide spread of intensities across the histogram.
- A low-contrast image has most intensities concentrated in a small range.

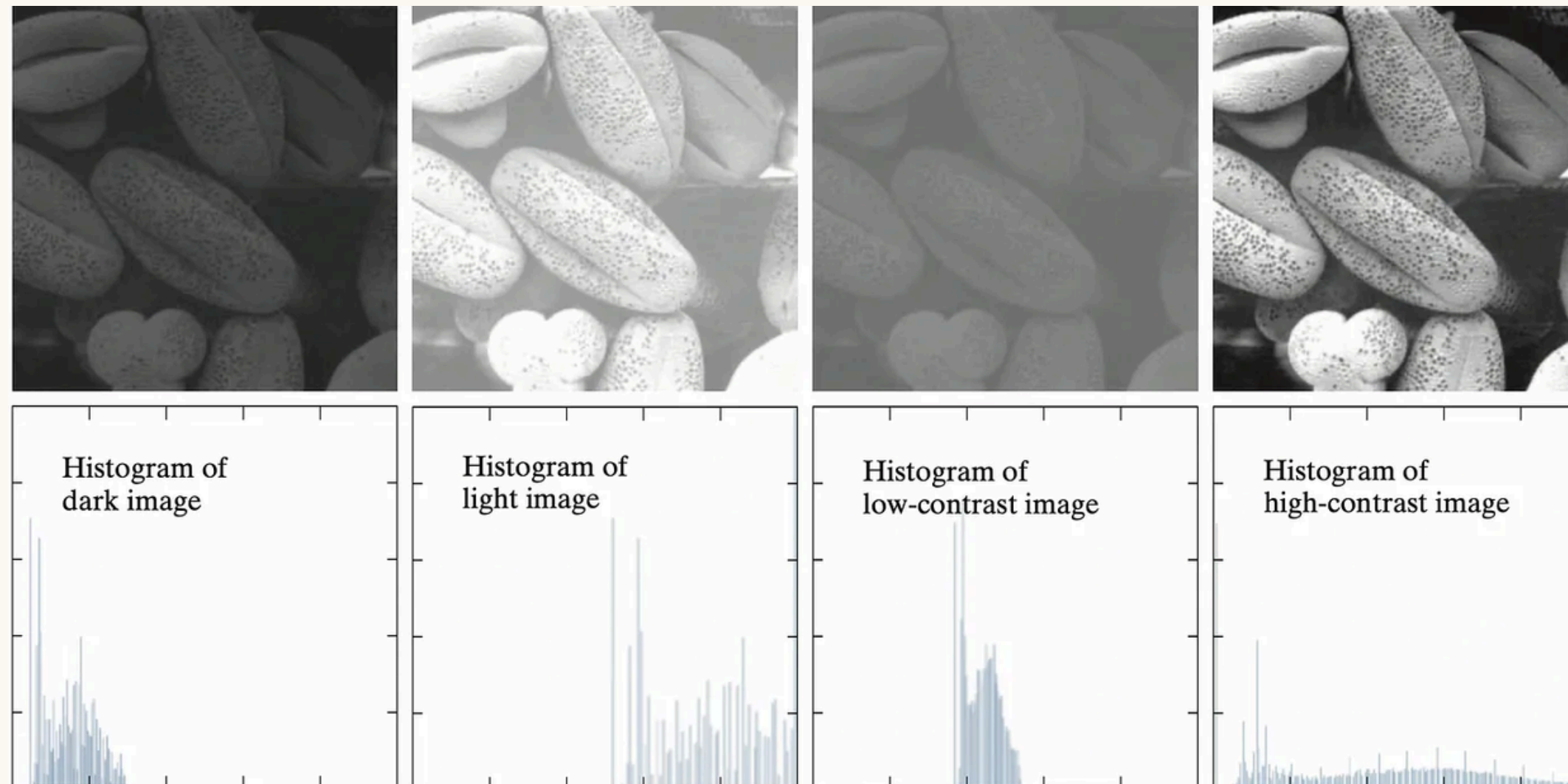


Image Analysis

Provides a clear visualization of the distribution of colors or brightness in an image, Helps identify overexposed (too bright) or underexposed (too dark) areas, aiding in image evaluation.

Image Processing

Facilitates the detection of features in an image, Enables the isolation of specific regions based on their color or brightness distribution.

Histogram benefits

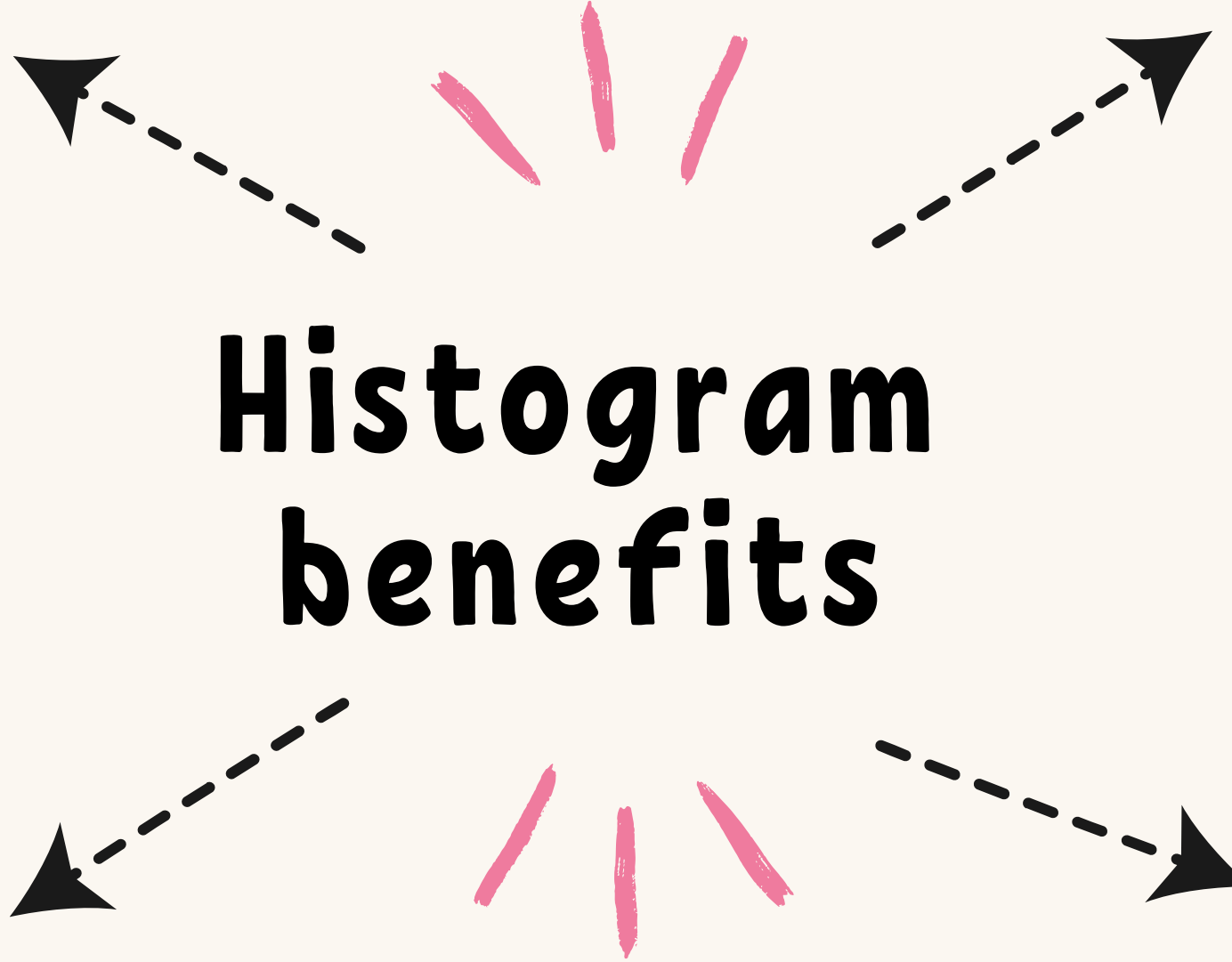
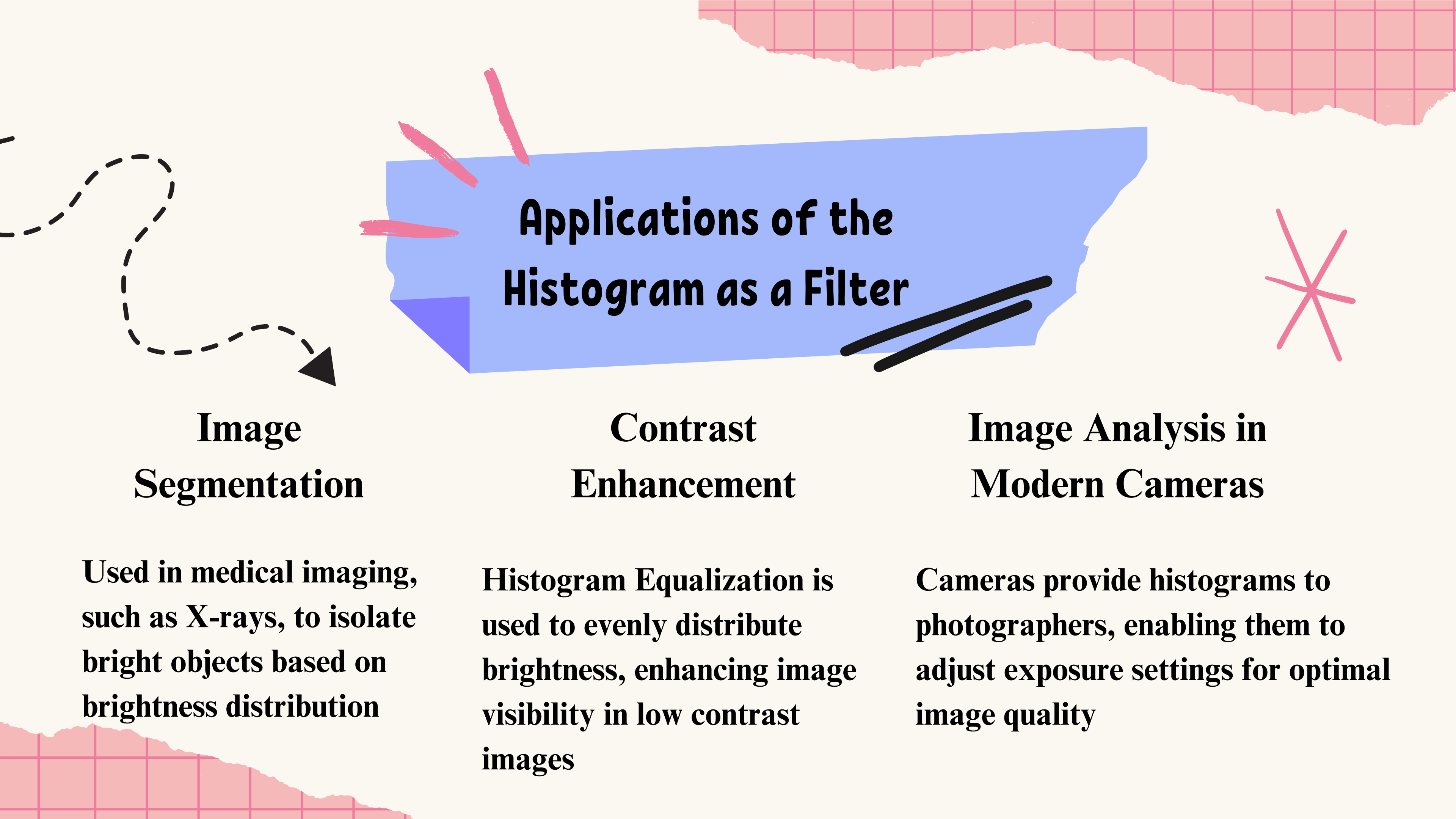


Image Filtering:

Thresholding: Converts an image into a binary (black and white) format, simplifying object detection and further processing tasks

Image Quality Enhancement

Histogram Equalization: Enhances contrast by redistributing intensity levels, making the image more balanced and visually appealing.



Applications of the Histogram as a Filter

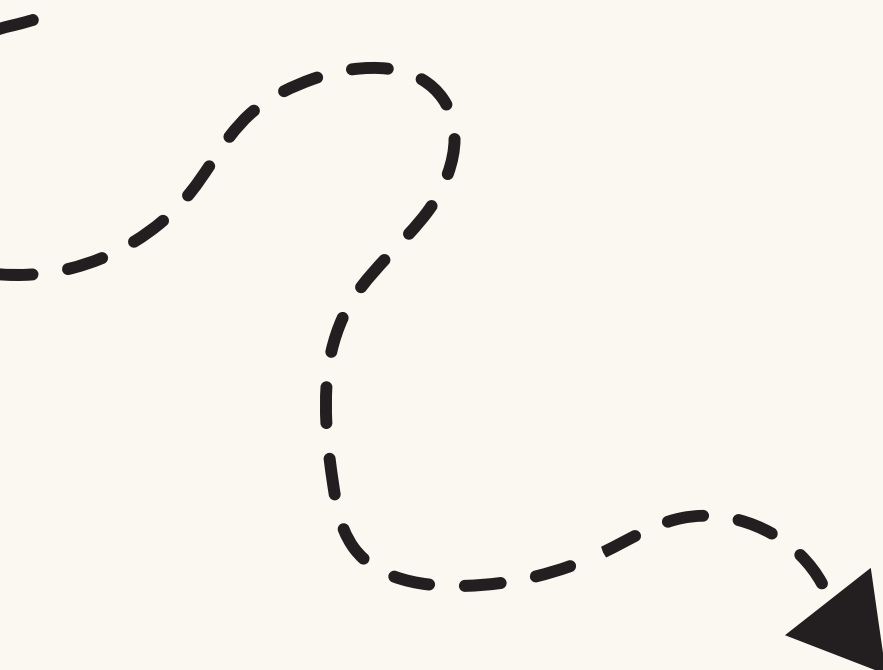


Image Segmentation

Used in medical imaging, such as X-rays, to isolate bright objects based on brightness distribution

Contrast Enhancement

Histogram Equalization is used to evenly distribute brightness, enhancing image visibility in low contrast images

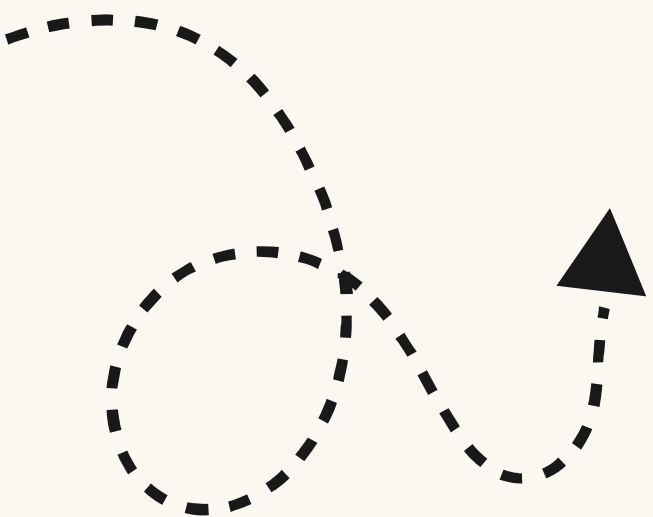
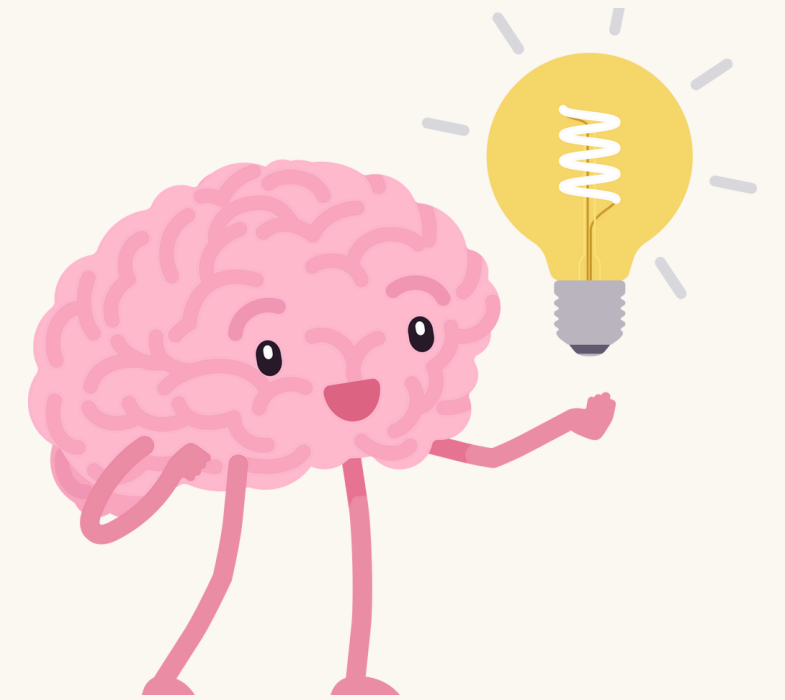
Image Analysis in Modern Cameras

Cameras provide histograms to photographers, enabling them to adjust exposure settings for optimal image quality

Let's take some examples

EX.1 :

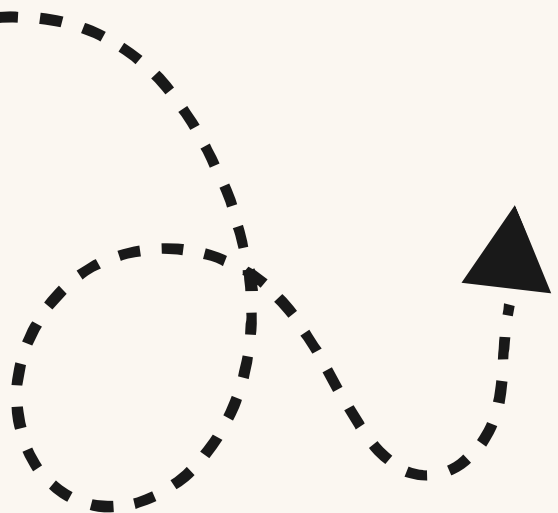
**suppose that you have gray image, find
Histogram for these intensity values [0, 50,
50, 100, 100, 150, 200, 255]**





Solution

pixel intensity	Freq (n_k)
0	1
50	2
100	2
150	1
200	1
255	1



Let's take some examples

EX.2 :

suppose that you have colorful image with these intensity values, find Histogram

1)(R): [100, 150, 200]

2)(G): [50, 100, 150]

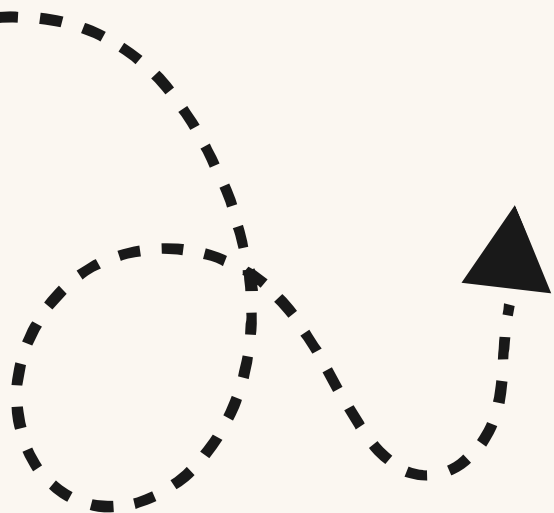
3)(B): [200, 150, 100]



Solution

For **RED** channel

pixel intensity	Freq (n_k)
100	1
150	1
200	1



Solution

For **GREEN** channel

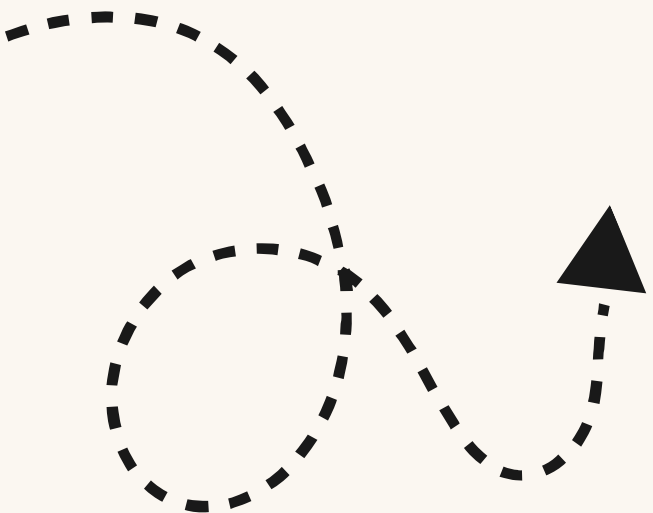
pixel intensity	Freq (n_k)
50	1
100	1
150	1



Solution

For **BLUE** channel

pixel intensity	Freq (n_k)
200	1
150	1
100	1



Let's take some examples

EX.3 :

suppose that you have this image with these intensity values , find Histogram

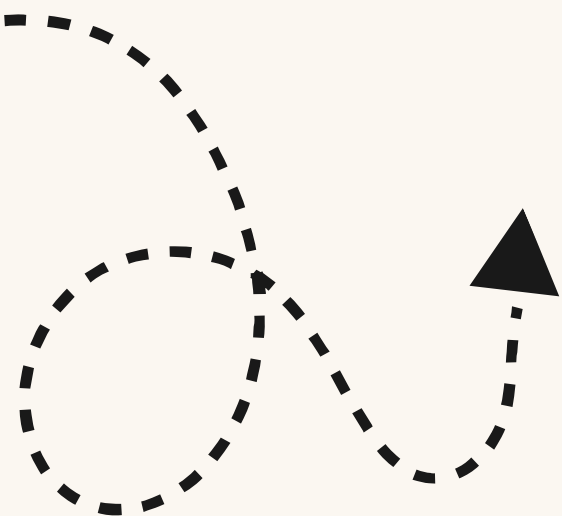
2	3	4	4	6
1	2	4	5	6
1	1	5	6	6
0	1	3	3	4
0	1	2	3	4



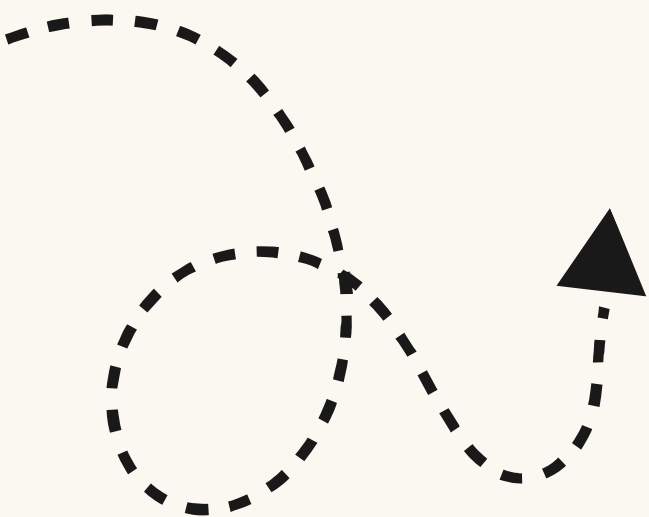
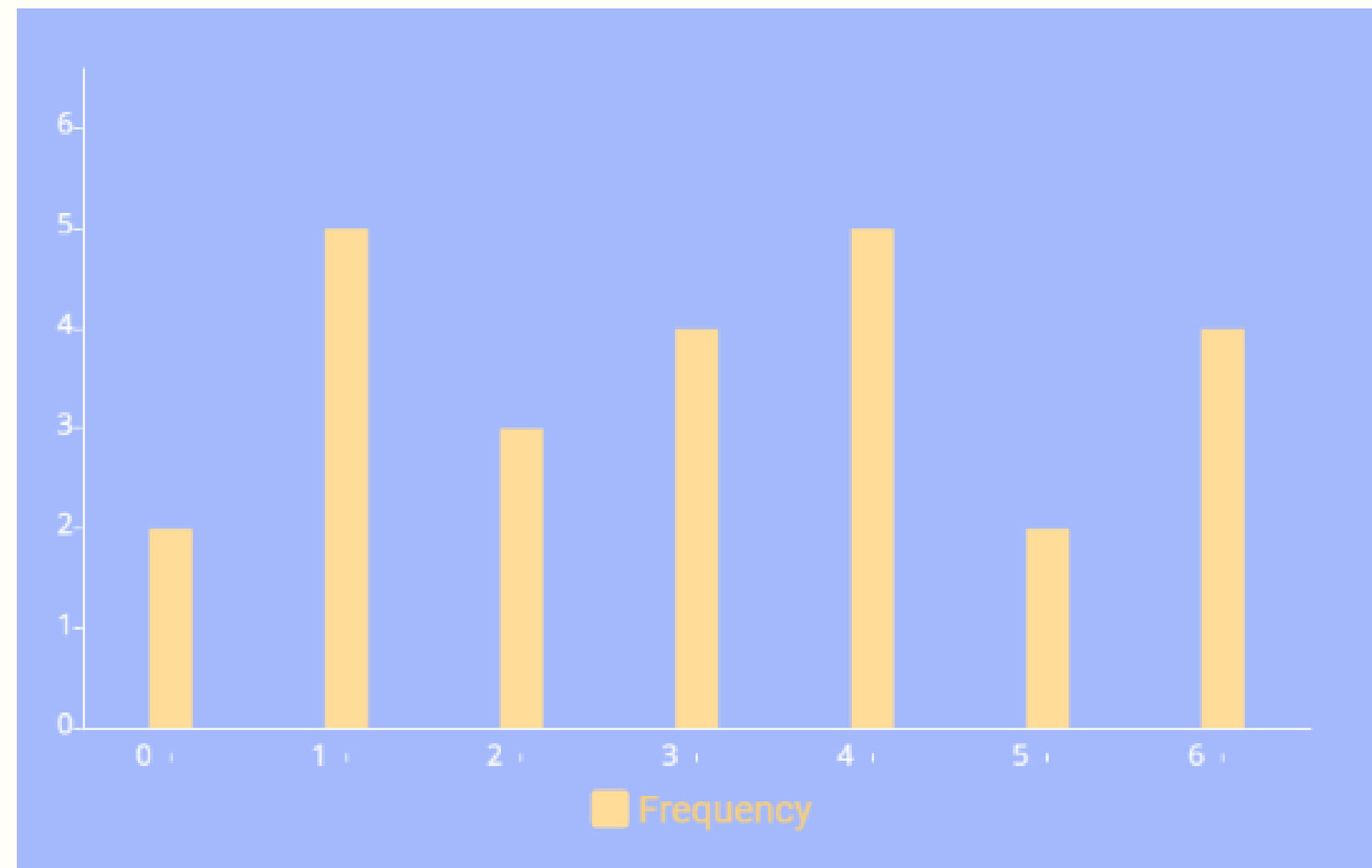


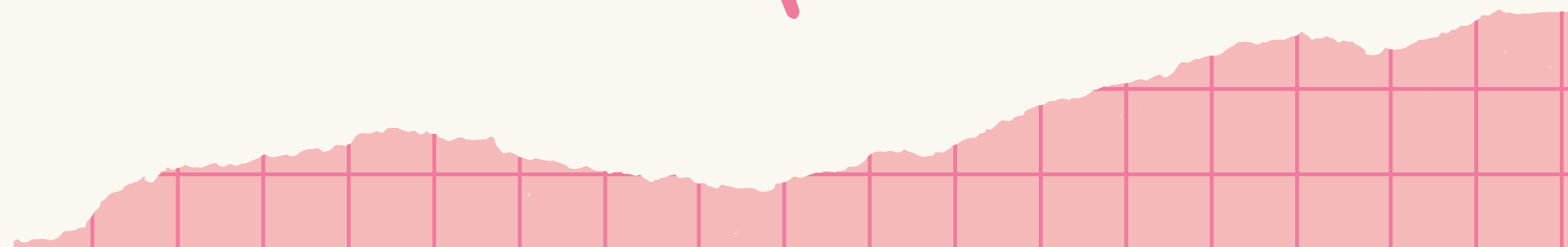
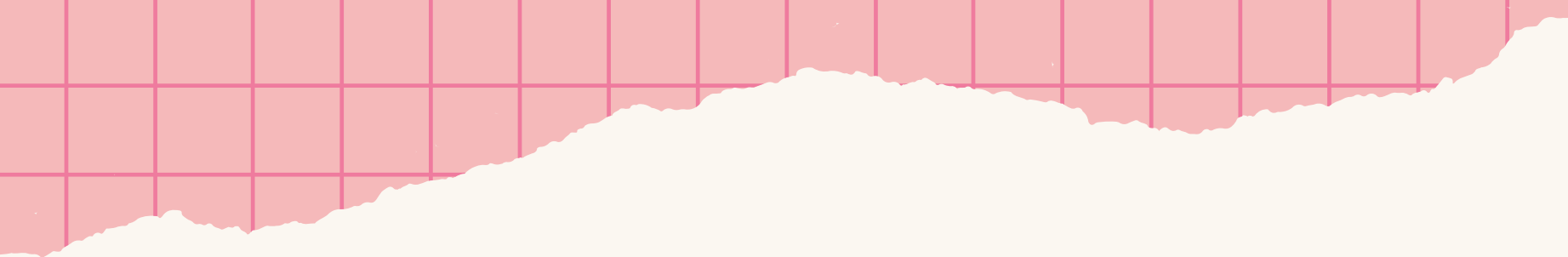
Solution

pixel intensity	Freq (n_k)
0	2
1	5
2	3
3	4
4	5
5	2
6	4

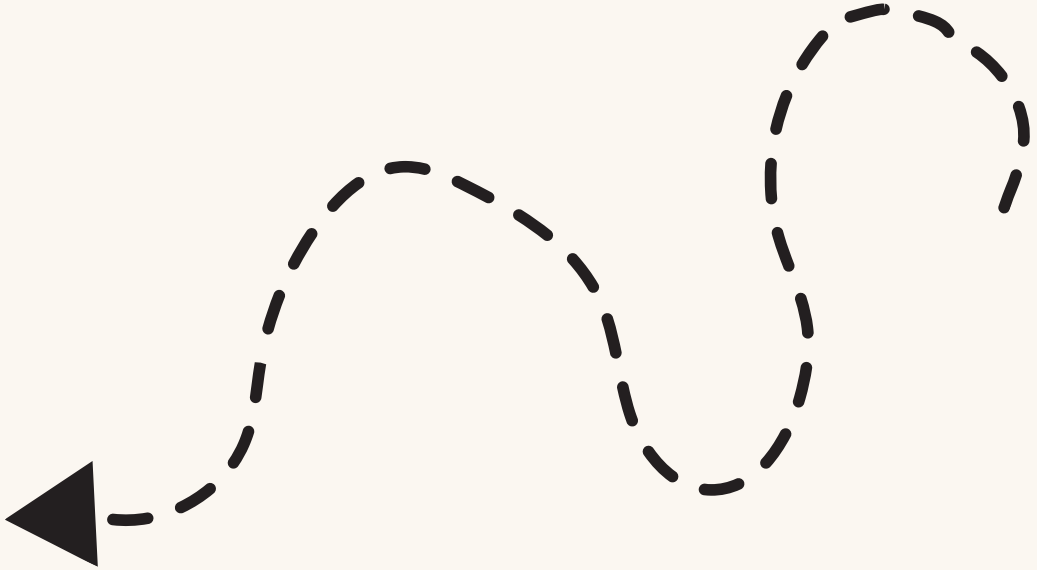


Solution





Normalized Histogram



What is a normalized histogram?

is a technique consisting into transforming the discrete distribution of intensities into a discrete distribution of probabilities. To do, we need to divide each value of the histogram by the number of pixel.

$$P(r_k) = h(r_k) / M*N$$

Where r_k is the k^{th} intensity value, $M*N$ is the number of pixels in the histogram

$h(r_k)$ is the number of pixels with intensity r

Let's take some examples

Ex.1

Suppose that you have 3-bit image ($L=8$) of size $64*64$ pixels has the gray level (intensity) distribution

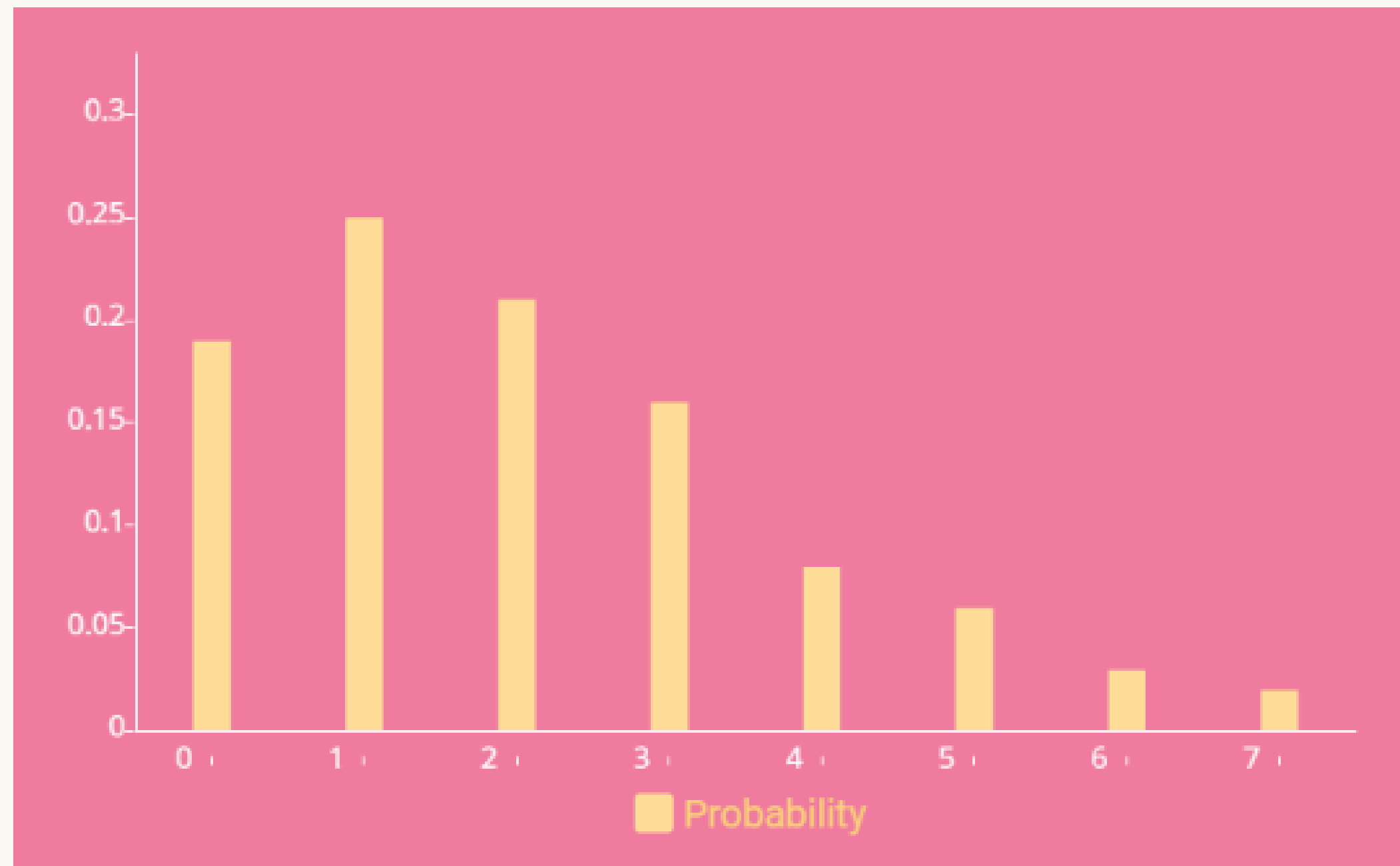
$n_k=(790,1023,850,656,329,245,122,81)$



Solution

$r(k)$	$h(r_k)$	$p(r_k) = h(r_k)/(M \times N)$
0	790	$790/4096 = 0.19$
1	1023	$1023/4096 = 0.25$
2	850	$850/4096 = 0.21$
3	656	$656/4096 = 0.16$
4	329	$329/4096 = 0.08$
5	245	$245/4096 = 0.06$
6	122	$122/4096 = 0.03$
7	81	$81/4096 = 0.02$
Total	4096	Almost = 1

Solution



Let's take some examples

Ex.2

Suppose that you have this image

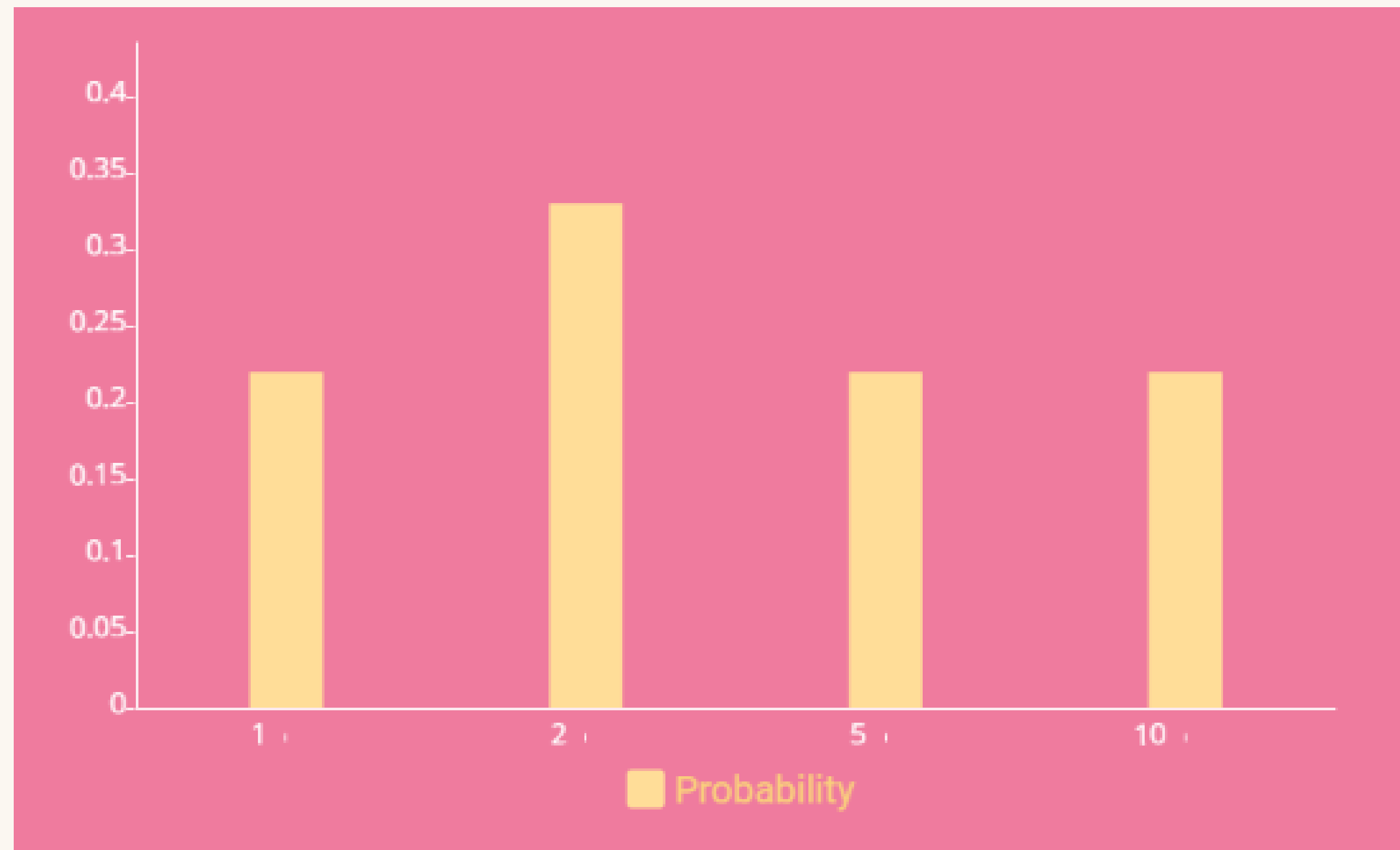
10	10	2
5	5	2
1	1	2



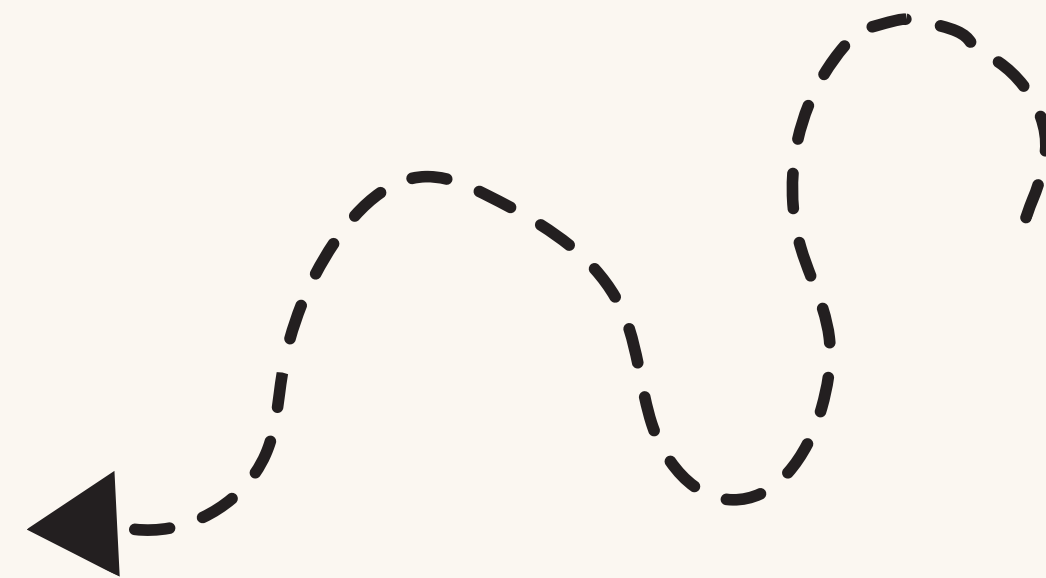
Solution

$r(k)$	$h(r_k)$	$p(r_k) = h(r_k)/(M \times N)$
1	2	$2/9 = 0.22$
2	3	$3/9 = 0.33$
5	2	$2/9 = 0.22$
10	2	$2/9 = 0.22$
Total	9	Almost = 1

Solution



Histogram Equalization

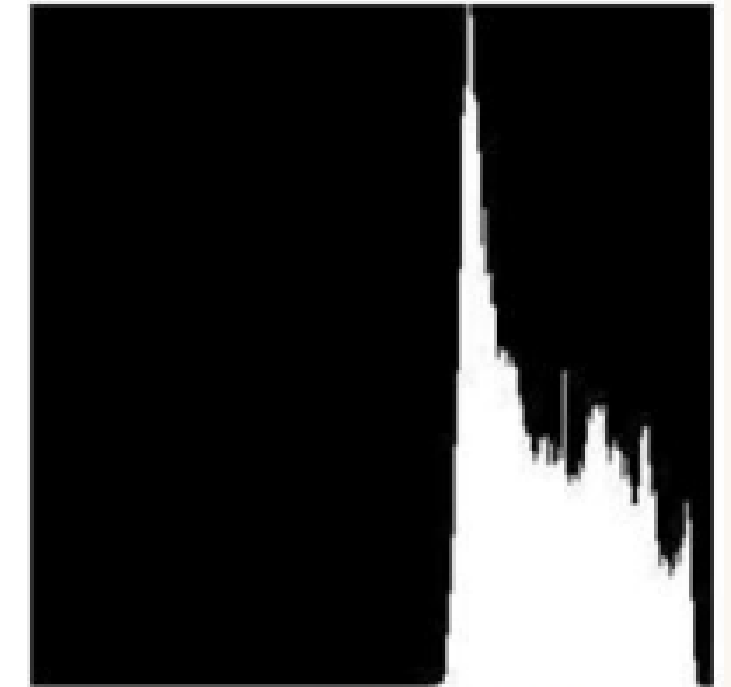


Histogram Equalization

Is a popular technique used for **improving the appearance** of a poor image.



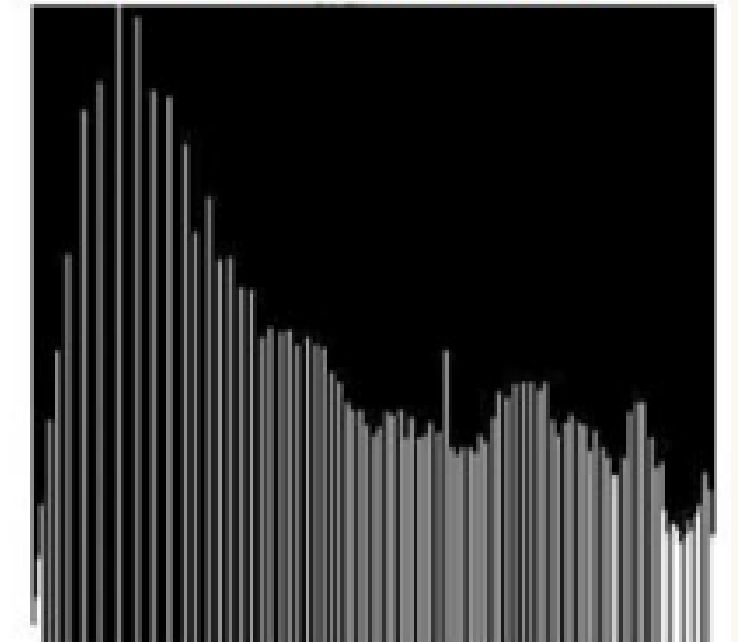
Original Image



Histogram of original image



Image after histogram equalization



Histogram after equalization



The histogram equalization process for digital images consists of four steps

Four Steps

Gray-level value	0	1	2	3	4	5	6	7
No of Pixel	10	8	9	2	14	1	5	2
Run sum	10	18	27	29	43	44	49	51

1

Find the running sum of the histogram values

Great a running sum of histogram values.

This means that the first values is 10,

the second is $10+8=18$,

next is $10+8+9=27$,

and soon.

Here we get 10,18,27,29,43,44,49,51.

Four Steps

2

Normalize the values from **step1** by dividing by **total number of pixels**

Normalize by dividing by total number of pixels.

The **total number of pixels** is

$$10+8+9+2+14+1+5+2=51.$$

This means that the **first values** is $10/51$,
the **second** is $18/51$,
next is $27/51$,
and soon.

Here we get $10/51, 18/51, 27/51, 29/51, 43/51, 44/51, 49/51, 51/51$.

Gray-level value	0	1	2	3	4	5	6	7
No of Pixel	10	8	9	2	14	1	5	2
Run sum	10	18	27	29	43	44	49	51
Normalized	$10/51$	$18/51$	$27/51$	$29/51$	$43/51$	$44/51$	$49/51$	$51/51$

Four Steps

We have an image with 3 bit /pixel, so the possible range of values is 0 to 7. $L=2^3=8$

Gray-level value	0	1	2	3	4	5	6	7
No of Pixel	10	8	9	2	14	1	5	2
Run sum	10	18	27	29	43	44	49	51
Normalized	$\frac{10}{51}$	$\frac{18}{51}$	$\frac{27}{51}$	$\frac{29}{51}$	$\frac{43}{51}$	$\frac{44}{51}$	$\frac{49}{51}$	$\frac{51}{51}$
Multiply by 7	1	2	4	4	6	6	7	7

3

Multiply the values from step2 by the $(L-1)$

Multiply these values by the $(L-1)$,
and then round the result to the closet integer.

This means that the first values is
 $(10/51)*7=1.3 \approx 1$,

the second is $(18/51)*7=2.4 \approx 2$,

next is $(27/51)*7=3.7 \approx 4$,

and soon.

After this is done we obtain 1,2,4,4,6,6,7,7.

Four Steps

Gray-level value	0	1	2	3	4	5	6	7
No of Pixel	10	8	9	2	14	1	5	2
Run sum	10	18	27	29	43	44	49	51
Normalized	$\frac{10}{51}$	$\frac{18}{51}$	$\frac{27}{51}$	$\frac{29}{51}$	$\frac{43}{51}$	$\frac{44}{51}$	$\frac{49}{51}$	$\frac{51}{51}$
Multiply by 7	1	2	4	4	6	6	7	7
New Value	1	2	4	4	6	6	7	7

- 4 Map the gray-level values to the results from step 3, using a one-to-one correspondence

Map the original values to the results from step 3 by a one-to-one correspondence.

هنشيل كل 0 في القيم القديمة ونحط 1
كقيمة جديدة بعد تطبيق histogram Equalization .

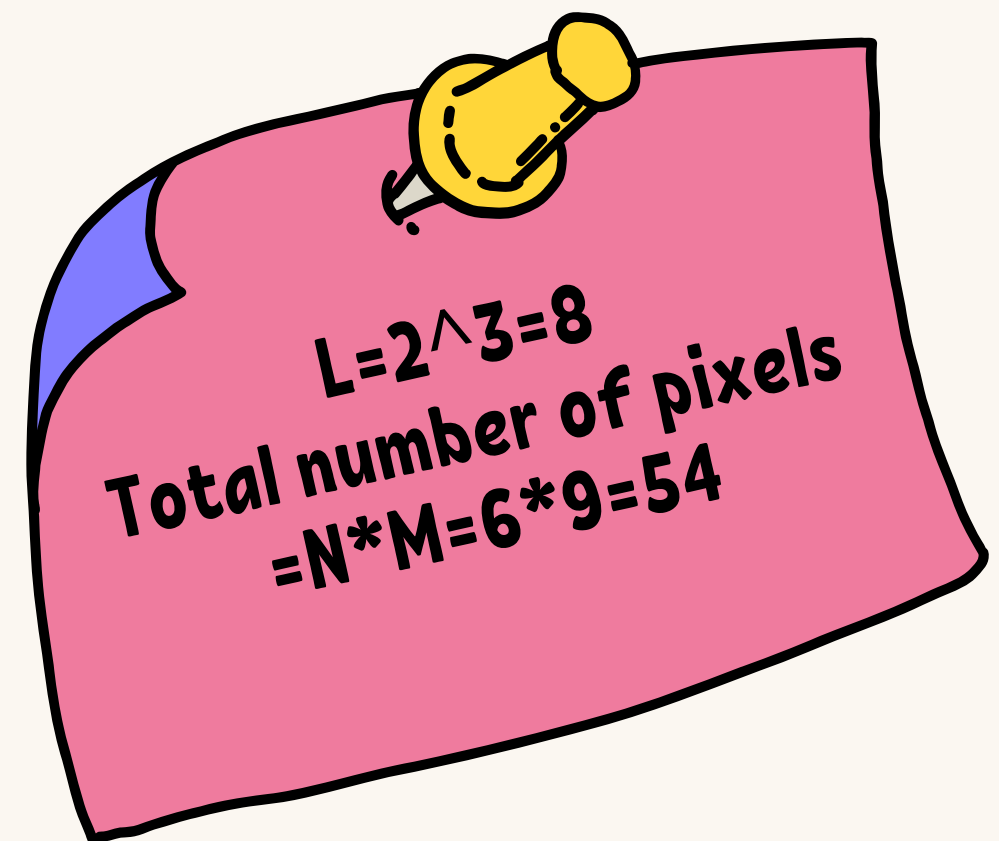
• New histogram After Apply histogram Eq.

New value	0	1	2	3	4	5	6	7
No of Pixel	0	10	8	0	11	0	15	7

Let's take some examples

Q/ Apply histogram equalization technique for the following (3-bit) sub image?

0	3	1	4	0	4	1	0	5
2	0	4	1	1	2	4	4	7
0	4	0	4	4	1	2	6	7
5	2	4	0	4	7	6	6	5
2	0	2	4	0	4	2	2	6
0	1	1	0	2	0	3	1	6



Solution

Convert matrix to histogram

Gray-level value	0	1	2	3	4	5	6	7
No of Pixel	12	8	9	2	12	3	5	3

1

Find the **running sum** of the **histogram values**

2

Normalize the values from **step1** by dividing by **total number of pixels**

3

Multiply the values from **step2** by the **(L-1)**

4

Map the **gray-level values** to the **results** from step 3, using a one-to-one correspondence

Solution

Gray-level value	0	1	2	3	4	5	6	7
No of Pixel	12	8	9	2	12	3	5	3
Run Sum	12	20	29	31	43	46	51	54
Normalized	12/ 54	20/ 54	29/ 54	31/ 54	43/ 54	46/ 54	51/ 54	54/ 54
Multiply by 7	2	3	4	4	6	6	7	7

Run Sum
first > 12
second > $12+8=20$
next > $12+8+9=29$...

Normalized
total no. of pixels=54
first > $12/54$
second > $20/54$
next > $29/54$...

Multiply by 7
first > $(12/54)*7=1.55 \approx 2$
second > $(20/54)*7=2.59 \approx 3$
next > $(29/54)*7=3.7 \approx 4$...

Solution

Step 4:

Old value	0	1	2	3	4	5	6	7
New value	2	3	4	4	6	6	7	7

$$\begin{bmatrix} 0 & 3 & 1 & 4 & 0 & 4 & 1 & 0 & 5 \\ 2 & 0 & 4 & 1 & 1 & 2 & 4 & 4 & 7 \\ 0 & 4 & 0 & 4 & 4 & 1 & 2 & 6 & 7 \\ 5 & 2 & 4 & 0 & 4 & 7 & 6 & 6 & 5 \\ 2 & 0 & 2 & 4 & 0 & 4 & 2 & 2 & 6 \\ 0 & 1 & 1 & 0 & 2 & 0 & 3 & 1 & 6 \end{bmatrix}$$

Original Image

$$\begin{pmatrix} 2 & 4 & 3 & 6 & 2 & 6 & 3 & 2 & 6 \\ 4 & 2 & 6 & 3 & 3 & 4 & 6 & 6 & 7 \\ 2 & 6 & 2 & 6 & 6 & 3 & 4 & 7 & 7 \\ 6 & 4 & 6 & 2 & 6 & 7 & 7 & 7 & 6 \\ 4 & 2 & 4 & 6 & 2 & 6 & 4 & 4 & 7 \\ 2 & 3 & 3 & 2 & 4 & 2 & 4 & 3 & 7 \end{pmatrix}$$

New image ,After Histogram Equalization

Question :(type MCQ)

Consider the following **5*5 image array** to answer the following question.
what is the value **after histogram equalization** at location (3,2) ?

a) 2 b) 3 c) 4 d) 5

Correct Answer : C) 4

3	3	3	6	4
2	2	0	0	4
1	2	4	0	5
5	1	3	2	1
5	2	6	6	6

total no of pixels = $5*5=25$
 $L-1=6$

Solution

Convert matrix to histogram

Gray-level value	0	1	2	3	4	5	6
No of Pixel	3	3	5	4	3	3	4

1

Find the **running sum** of the **histogram values**

Run Sum
 $3+3+5+4=15$

2

Normalize the values from **step1** by dividing by **total number of pixels**

Normalized
 $15/25$

3

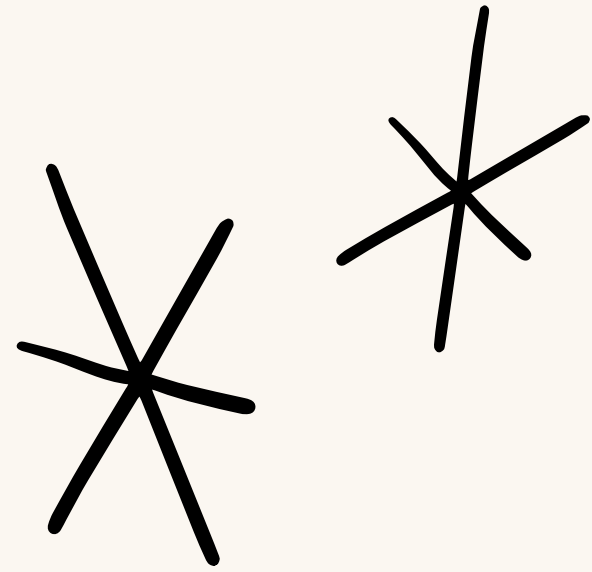
Multiply the values from **step2** by the **(L-1)**

Multiply by 6
 $(15/25)*6=3.6 \approx 4$

4

Map the **gray-level values** to the **results** from step 3, using a one-to-one correspondence

old value 3
new value 4
Result 4 ✓



Our team

Mustafa Mohamed Mustafa

Hossam Ahmed Mohamed

Mohamed Ahmed Salama

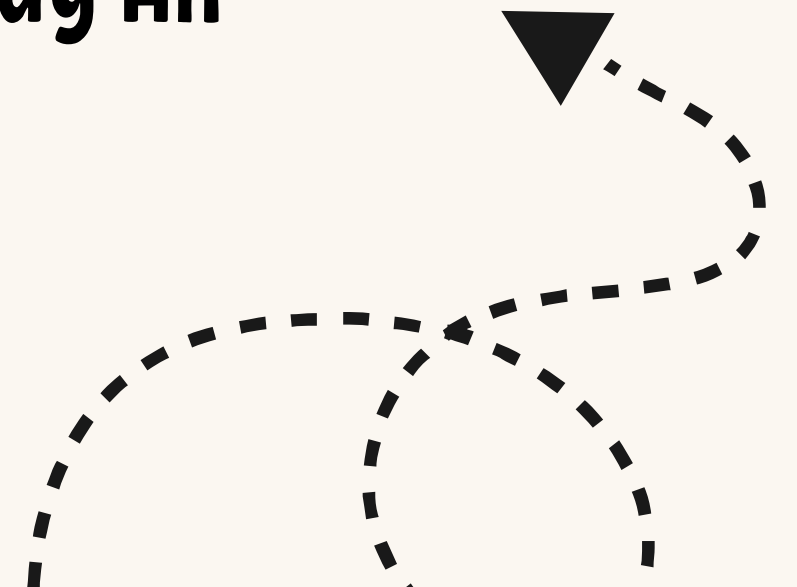
Ahmed Hassan Gamal

Doaa Abdallah El-Sayed

Sama Saber Abdel-aziz

Eslam Salah El-Sayed

Zeinab Hamdy Ali



THANK
YOU!

